

**HINTS & SOLUTIONS**

Q.1 (C)

$$q = w$$

$$\Delta U = q + w$$

$$\Delta U = q + q = 2q$$

$$nC_{vm}\Delta T = 2 \times n \times C_m \Delta T$$

$$\text{or } C_m = \frac{C_{vm}}{2}$$

$$\text{or } C_m = \frac{R}{(\gamma - 1) \times 2} = R$$

Q.2 (D)

Q.3 (A)

Q.4 (C)

$$q = 0$$

$$\Delta U = W = -P_{\text{ext}}(V_2 - V_1)$$

$$= -4(40 - 50) = +40 \text{ bar-L}$$

$$\Delta H = \Delta U + \Delta(PV)$$

$$= +40 + (P_2V_2 - P_1V_1) \text{ bar-L}$$

$$= +40 + (200 - 150) \text{ bar-L}$$

$$= 90 \text{ bar-L} = 9000 \text{ J}$$

Q.5 (C)

Q.6 (C)

Q.7 (C)



for tetramerization of liquid

$$\Delta H = 30 \times 4 - 100 - 72$$

$$\Delta H = -52 \text{ kJ/mol}$$

$$\Delta S = -125 \text{ J/mol / K}$$

$$\Delta G = \Delta H - T\Delta S$$

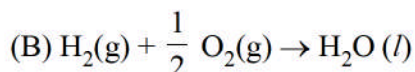
$$\Delta G = -52 - 300 \times (-125) / 1000 = -52 + 37.5$$

$$\Delta G = -14.5 \text{ kJ/mol}$$

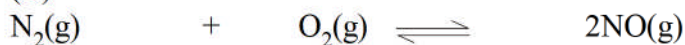


Q.8 (B)

For the reaction to be non-spontaneous at high temperature $\Delta S = -ve$, $\Delta H = -ve$



Q.9 (C)



$$\Delta G = \Delta G^\circ + RT \ln Q$$

$$\Delta G = [\Delta H^\circ - T\Delta S^\circ] + RT \ln Q$$

$$\Delta H^\circ_{Rxn} = 2\Delta H^\circ_f NO = 90.5 \times 2 \text{ kJ}$$

$$\Delta S^\circ_{Rxn} = 2\Delta S^\circ_f NO = 12.5 \times 2 \text{ J/K}$$

$$Q = \frac{P_{NO}^2}{P_{N_2} \times P_{O_2}} = \frac{10^{-6}}{10^{-1} \times 10^{-3}} = 10^{-2}$$

$$\Delta G = \left[90.5 \times 2 \text{ kJ} - \frac{300 \times 12.5 \times 2}{1000} \right] + \frac{2.303 \times R \times 300 \times \log 10^{-2}}{1000}$$

$$\Delta G = 162 \text{ kJ/mole}$$

Q.10 (B)

Entropy of universe isolated system always increases in a spontaneous process so statement - I is incorrect.

Q.11 (B)



(B) During polymerisation $\Delta S \downarrow$ due to chain formation.

$$(C) \Delta S = nC_{v,m} \ln \frac{T_f}{T_i} + nR \ln \frac{V_f}{V_i} \text{ for irreversible compression } (T_f)_{irr.} > (T_f)_{rev.}$$

$$\Rightarrow \Delta S > 0 \quad \therefore \text{for reversible } \Delta S = 0$$

$$(D) \Delta S = nR \ln \frac{V_f}{V_i} \text{ for free expansion } (V_f > V_i)$$

Q.12 (C)

$$\Delta H^\circ_{vap} = 3 \times 40 = 120 \text{ kJ/mole}$$

$$\Delta S^\circ_{vap} = \frac{120 \text{ kJ}}{300 \text{ mole-K}} = 400 \frac{\text{J}}{\text{mole-K}}$$

$$\Delta S^\circ_{condensation} = -400 \frac{\text{J}}{\text{mole-K}}$$

Q.13 (C)

For diatomic ideal gas $C_{v,m} = \frac{5}{2}R$

Given : $T \propto V^3$

$$T = KV^3$$

$$\frac{PV}{nR} = KV^3$$

$$PV^{-2} = K$$

For polytropic process

$$PV^x = K$$

$$\begin{aligned} C_m &= C_{v,m} + \frac{R}{(1-x)} \\ &= \frac{5}{2}R + \frac{R}{1-(-2)} \\ &= \frac{5}{2}R + \frac{R}{3} \\ &= \frac{15R + 2R}{6} = \frac{17R}{6} \end{aligned}$$

Q.14 (D)

Expansion is adiabatic & free

$$q = 0$$

$$w = 0$$

$$\Delta U = q + w$$

$\Delta U = 0 \Rightarrow$ No change of internal energy.

So change in internal energy will be zero.

But for real gas or other substance it is not necessary that $\Delta T = 0$

Because $U = f(P, V, T)$ for others.

Q.15 (D)

Expansion is adiabatic & free

$$q = 0$$

$$w = 0$$

$$\Delta U = q + w$$

$\Delta U = 0 \Rightarrow$ No change of internal energy.

So change in internal energy will be zero.

But for real gas or other substance it is not necessary that $\Delta T = 0$

Because $U = f(P, V, T)$ for others.

Q.16 (A)(D)

a process is spontaneous

$$(\Delta G)_{P,T} < 0$$

$$(\Delta H - T\Delta S)_{P,T} < 0$$

$$(\Delta A)_{V,T} < 0$$

$$(dU - TdS)_{V,T} < 0$$

(A) $P = \text{constant}$

$$\Delta S = 0$$

$(\Delta H)_{P,S} < 0$ Process must be spontaneous

(C) $dU = 0$ Process will be spontaneous only if it occurs at constant volume.

(D) $TdS > dQ$

$dS > 0$ process is spontaneous

Q.17 (A) (B) (D)

$$\Delta G = \Delta H - T\Delta S$$

(i) Combustion is spontaneous reaction so ΔH is -ve as $\Delta H > T\Delta S$

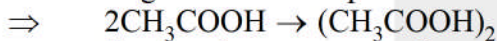
(ii) ΔG is -ve for vaporisation above boiling point as $T\Delta S > \Delta H$

(iii) ΔG is +ve for fusion at 0°C as $\Delta H = +\text{ve}$ and $\Delta S = -\text{ve}$

(iv) ΔG is -ve as $\Delta S = +\text{ve}$ & $\Delta H = +\text{ve}$ but $T\Delta S > \Delta H$

Q.18 (A) (B) (C) (D)

$\Rightarrow$ Combustion of hydrocarbon is always an exothermic process. so if the process is occurring in an adiabatic rigid container temperature must increase.



Entropy decreases as number of moles decreases.

$\Rightarrow$ In adiabatic process, between any two states there can be only one path.

$\Rightarrow$ For a reaction involving ideal gas only.

$R \rightarrow P$ at constant T

$$\Delta U = (U)_P - (U)_R \neq 0$$

$\Rightarrow$ In case of free expansion $P_{\text{ext}} = 0$

$\therefore W = 0$ even if there is change in volume.

Q.19 (A) (B) (C) (D)

$\therefore$ Ideal gas & initial & final temperature are same

$\therefore \Delta E = 0$ for all

$\therefore$ (C) is **correct**

Also $\therefore |w|$ is maximum for SO_2 is step I & overall

$\Delta E = 0 \therefore |q|$ for SO_2 will be maximum

$\therefore$ option (B) is **correct**

Since after step I, all have same volume, same moles & pressure of Ar is minimum $\therefore$ Temperature of Ar will be lowest

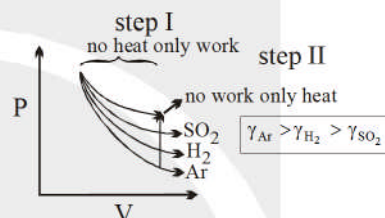
$\therefore$ decrease in temperature will be maximum

$\therefore$ option (A) is **correct**

$\therefore \gamma$ for Ar & SO_2 is different (due to vibrational degrees of freedom)

$\therefore$ graph is different

$\therefore$ option (D) is **correct**;



Q.20 -900 calories

$$n_1 C_{v1} dT + n_2 C_{v2} dT = -(n_1 RT + n_2 RT) \frac{dV}{V}$$

$$\Rightarrow \frac{dT}{T} = -\frac{R}{C_v} \frac{dV}{V}$$

$$\ln \frac{T_2}{T_1} = -\frac{1}{2} \ln \frac{V_2}{V_1}$$

$$\Rightarrow \frac{T_2}{T_1} = \left(\frac{V_1}{V_2} \right)^{1/2}$$

$$T_2 - T_1 = \frac{T_1}{2} - T_1 = -\frac{T_1}{2}$$

$$\Delta H = \frac{1}{2} \left(\frac{3}{2} R + R \right) + \frac{1}{2} \left(\frac{5}{2} R + R \right) (\Delta T)$$

$$= \frac{1}{2} (6R) \Delta T = 3R \Delta T = -3(2)(150) = -900 \text{ calories}$$



Q.21 2300

$$W = -P_{\text{ext}} (V_2 - V_1) = -3 \text{ Lbar.}$$

$$\frac{P_1 V_1}{T_1} = \frac{P_2 V_2}{T_2} \Rightarrow T_2 = 600 \text{ K}$$

$$\Delta U = nC_{V_m} \Delta T$$

$$= 20 \text{ L bar.}$$

$$= \Delta U = q + w$$

$$\Rightarrow q = 23 \text{ Lbar} = 2300 \text{ J}$$

Table-1			
State	P	V	T
1	1 atm	22.4	273
2	2 atm	22.4	546
3	1 atm	44.8	546

Q.22

Step	Name of process	q	w	ΔE	ΔH
A	Isochoric	$3/2 R(273)$	0	$3/2 R(273)$	$5/2 R(273)$
B	Isothermal	$546 R \ln 2$	$-546 R \ln 2$	0	0
C	Isobaric	$-5/2 R(273)$	$R(273)$	$-3/2 R(273)$	$-5/2 R(273)$

Q.23 $\Delta H \cong \Delta U = 1440 \text{ calories}$

Q.24 $\Delta U = 27.91 \text{ kJ mol}^{-1}$, $t = 514 \text{ sec.}$

Q.25 (i) $q = -w = 17.54 \text{ kJ}$, $\Delta U = 0$ and $\Delta H = 0$; (ii) $q = 0$, $w = \Delta U = -10.536 \text{ kJ}$ and $\Delta H = -14.75 \text{ kJ}$
 (iii) $q = 0$, $w = 0$, $\Delta U = 0$ and $\Delta H = 0$ (iv) $q = 0$; $\Delta U = w = -7.14 \text{ KJ}$; $\Delta H = -9.996 \text{ KJ}$,
 (v) $q = -w = 10.0 \text{ KJ}$, $\Delta U = \Delta H = 0$

Q.26 320

Both the gases before mixing are at same temperature and pressure. Since the amount of neon is twice that of Helium.

$$V_{\text{He}} = 2 V_{\text{He}}$$

$$V_{\text{total}} = V_{\text{He}} + V_{\text{Ne}} = V_{\text{He}} + 2V_{\text{He}} = 3 V_{\text{He}}$$

$$\Delta S_{\text{He}} = nR \ln \frac{V_2}{V_1} = 1 \times 2 \times \ln 3 = 2 \text{ Cal K}^{-1}$$

$$\Delta S_{\text{Ne}} = nR \ln \frac{V_2}{V_1} = 2 \times 2 \times \ln \frac{3}{2} = 1.2 \text{ Cal K}^{-1}$$

$$\Delta S_{\text{Total system}} = 3.2 \text{ cal K}^{-1}$$

$$\Delta S = 3.2 \times 100 = 320 \text{ Ans.}$$

Q.27 $\Delta H^\circ = 9.04 \text{ kJ/mol}$; $\Delta S^\circ = -8.64 \text{ J/mol}^{-1} \text{K}^{-1}$

Q.28 0400



$$\Delta H = \Delta U + \Delta(PV)$$

$$\Delta H = \Delta U + P(\Delta V)$$

$$-500 = \Delta U + 1 \times \frac{(-20) \times 50}{1000} \times 100$$

$$\Delta U = -400 \text{ J/Mole}$$

$$|\Delta U| = 400$$

Q.29 0008

CD $\rightarrow$ is isothermal

DA $\rightarrow$ is isobaric

$$T_A > T_D$$

For BC

PV both decrease

T also decrease

For AB

$$P - 1 = \frac{3 - 1}{2 - 6}(V - 6)$$

$$PV = RT$$

$$P - 1 = -\frac{V}{2} + 3$$

$$T = \frac{RT}{V}$$

$$P = -\frac{V}{2} + 4$$

$$RT = -\frac{V^2}{2} + 4V$$

$$R \frac{dT}{dV} - V + 4 = 0$$

$$V = 4L$$

$$PV = RT$$

$$T_{\max} = -\frac{16}{2} + 16 = +\frac{8}{R}$$

